Latent Bridge Matching: IDT-Calibrated Latent Transport for Domain-Level Artistic Transfer

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Abstract

Domain-level artistic transfer has a blunt evaluation failure: when the source is already art, doing nothing can score as target style. On the CLIP-separated WikiArt Distinct5 stress split, the reproduced SaMAM run remains below the identical-image transfer-only CLIP-S floor through its last trusted 2250-step checkpoint. We introduce *Latent Bridge Matching* (LBM), a 3.9M-parameter latent-transport model that uses training-side endpoint pairing, a style-conditioned residual field, and terminal distribution matching in VAE latent space, keeping style-specific customization in a small-model learning regime rather than adapting a large generative prior. On Distinct5-512, the reviewed LBM-F/K operating points exceed IDT by +0.0244/+0.0312 transfer-only CLIP-S at the first reproduced RTX 3060 checkpoint (1.2 minutes), whereas SaMST exceeds the same floor only in a much higher-displacement, higher-targetwise-ArtFID region. On the historical strict-750 reference surface, LBM reaches 0.7161 CLIP-S / 0.4514 LPIPS at 114 ms/image. The practical reading is therefore not only metric-side caution but also accessibility: on this stress split, checking the unchanged-image floor prevents false stylization claims, while the reproduced minute-scale consumer-GPU path shows that customized style modeling can remain a lightweight learning problem for researchers and creators without large-scale compute.

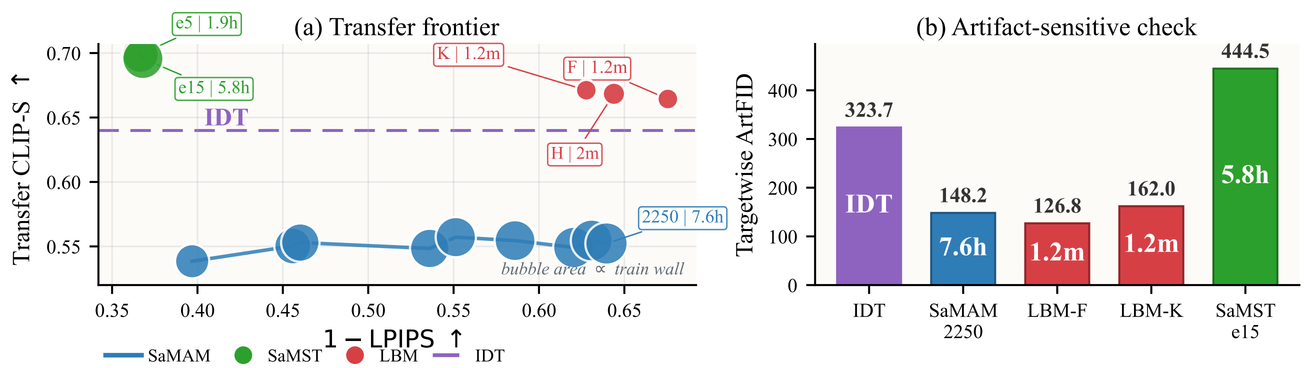


Figure 1: IDT-calibrated Distinct5-512 summary under the shared full 5×5 / 750-output packet. In (a), bubble area is cumulative training wall time with evaluation excluded. LBM reaches positive transfer-only gain above IDT at minute-scale cost, whereas the trusted SaMAM-2250 checkpoint remains below IDT and SaMST exceeds IDT only in a much higher-displacement regime. In (b), targetwise ArtFID shows that SaMST's stronger raw style is accompanied by a severe artifact penalty.

Introduction

Domain-level artistic transfer asks for a stylized image from a source image and a target style identifier. This is not only an image-generation problem but also a customization problem: the model should not need a reference style image or per-image optimization at inference, and the adaptation path should remain feasible without maintaining a large generative backbone. In practice, three pressures appear together. First, evaluation can fail badly when both source and target domains are artworks: an unchanged source can already obtain a strong target-style score. A method can therefore appear competitive by preserving an art-domain prior, or by making visible but misdirected edits, without moving toward the requested target domain. The first question should be signed target-style movement over the unchanged-image baseline, not raw target-style affinity.

Second, many visually strong stylization pipelines now rely on exemplar-conditioned correspondence or adaptation/steering of large pretrained generators. Those systems can be compelling, but they are awkward if the goal is to build a lightweight style-id model that a small lab, researcher, or creator can retrain and deploy at low cost. For the style-id setting studied here, the practical question is therefore not only *whether* target-style movement happens, but also *who can afford to customize the model and under what adaptation budget*. A useful regime should shift customized stylization from a resource-intensive adaptation routine toward minute-scale learning on consumer GPUs rather than multi-hour tuning on specialized setups.

This failure is not the familiar style/content trade-off. Classical neural style transfer introduced feature reconstruction and Gram statistics (Gatys, Ecker, and Bethge 2016); feed-forward and attention-based arbitrary style transfer improved efficiency and local correspondence (Huang and Belongie 2017; Park and Lee 2019; Deng et al. 2022; Zheng et al. 2024). Recent systems now span reference-guided stylization (Hong et al. 2023; Huang et al. 2024; Zhang et al. 2025; Shang et al. 2025), compact multi-style models selected by style identity (Liu et al. 2024), and diffusion-prior stylizers ranging from trainable prompt-bank adaptation to training-free attention control (Zhang et al. 2024; Chung, Hyun, and Heo 2024; Deng et al. 2024; Jiang et al. 2025; Xu et al. 2025). Across these regimes, automatic metrics can conflate three different events: preserving a source that is already artistic, producing plausible art-domain texture, and moving toward the requested target style. In the style-id setting, the distinction is decisive because inference receives only a domain label.

The representation problem is geometric as well as statistical. VAE latents are compressed feature coordinates with nontrivial local curvature, not RGB pixels. Treating target style as pointwise Euclidean endpoint reconstruction can encourage mean solutions, suppress semantically valid local rearrangements, and blur the line between preserving content and doing nothing. We instead model stylization as controlled latent transport: target-domain examples select where supervision should pull, a style-conditioned vector field decides how to move, and terminal distribution matching checks whether the integrated endpoint has target-style patch statistics.

We propose *Latent Bridge Matching* (LBM), a compact latent-transport model built around that evaluation requirement. In the current Distinct5 mainline, LBM combines (i) training-side endpoint pairing instantiated through a prototype-aware pairing cache and endpoint-mode OMF training; (ii) a style-conditioned latent residual field that executes the requested transport; and (iii) SA-SWD with kinetic regularization, which pulls the Euler-integrated endpoint toward target-style patch statistics while limiting unnecessary displacement. A learnable style spatial prior and semantic cross-attention provide domain-level conditioning without per-image optimization or diffusion sampling. This keeps customization in a lightweight style-id regime: the reported model has 3.91M parameters, clears the Distinct5 IDT floor after 1.2 minutes of checkpoint training on an RTX 3060, and avoids adapting a large pretrained generator. In this paper, the identical-image floor is used as a split-local reporting check for the Distinct5 stress benchmark: style gain is read against that reference rather than from raw target-style affinity alone.

Our contributions are:

- **We formulate domain-level artistic transfer as executable latent transport**, separating style specification from latent execution through training-side endpoint pairing, time-conditioned field learning, explicit motion control, and terminal distribution matching.
- **We use the Identical-Image Transfer (IDT) floor as a split-local diagnostic on Distinct5-512**, where a reproduced baseline can visibly edit images yet still score below IDT in transfer-only CLIP-S, and we report Δ_{idt} as a companion metric that measures style movement beyond that floor.
- **We demonstrate a lightweight adaptation regime for customized style models**: in the reviewed setup, LBM clears the Distinct5-512 IDT floor after 1.2 minutes of checkpoint training on an RTX 3060, and the historical strict-750 model remains 3.91M with 114 ms/image inference. This lowers the barrier for researchers and creators to build customized style models without adapting a large generative prior.

Related Work

Classical and arbitrary neural style transfer. Neural style transfer began with optimization-based feature reconstruction and Gram-matrix style losses (Gatys, Ecker, and Bethge 2016). AdaIN later enabled real-time arbitrary transfer by aligning channel-wise feature statistics (Huang and Belongie 2017). Attention- and flow-based arbitrary style transfer then pushed local correspondence further: SANet improved style-aware matching, ArtFlow used reversible flows to reduce biased transfer, AdaAttN revisited local attentive normalization, StyTr2 introduced transformer-based content-style interaction, and S2WAT strengthened transformer locality with strip-window attention (Park and Lee 2019; An et al. 2021; Liu et al. 2021; Deng et al. 2022; Xia et al. 2024). More recent reference-guided arbitrary stylization continues to refine the injection path itself: AesPA-Net and AesFA add aesthetic-aware control, Puff-Net emphasizes cleaner content/style feature separation, HSI replaces point-wise attention with a holistic injector, and SCSA focuses on semantic-region consistency in arbitrary semantic style transfer (Hong et al. 2023; Huang et al. 2024; Zheng et al. 2024; Zhang et al. 2025; Shang et al. 2025). These works mainly ask how a style exemplar should be fused at inference time. Our problem is different: the main protocol in this paper does not provide a target style image at inference, so the central question is how a domain-level style control should be executed without exemplar-conditioned correspondence.

Efficient multi-style transfer and compact style representations. Multi-style transfer amortizes many styles into one compact model rather than solving a fresh arbitrary-style problem for every target image. SaMST is the closest recent representative: it separates style modeling from style transfer through pluggable style representations and a style-aware transfer network, explicitly targeting the quality-efficiency trade-off for many-style deployment (Liu et al. 2024). This line is the most relevant tokenizer-side baseline for our work. Throughout this paper, “style tokenizer” means a style-id-conditioned control map rather than an image tokenizer, a target-image encoder, or a diffusion-token controller. Our contribution is not another pluggable codebook alone. Instead, we ask whether a compact style representation remains executable after it passes through a latent transport field, so that style-side distinctions survive downstream rendering strongly enough to produce real style movement beyond the unchanged-image baseline.

State-space and Mamba-based style transfer. Recent work also revisits style transfer through state-space backbones. Mamba-ST and SaMam argue that linear-complexity state-space operators can recover global receptive fields more efficiently than quadratic attention, improving the efficiency-quality frontier of arbitrary or many-style transfer (Botti et al. 2025; Liu et al. 2025). These papers are important because they shift the architectural question within compact feed-forward arbitrary and multi-style stylizers from "CNN or transformer" to "which global-mixing primitive is most efficient." Our comparisons to SaMam take this line seriously. The distinction is that LBM does not claim a new global-mixing backbone; its main contribution is training-side and objective-side: endpoint pairing, transport-field execution, and semantic-aligned terminal distribution matching under a compact style-id interface.

Training-free and diffusion methods. Training-free and diffusion-based approaches provide a different route to flexible style transfer. CAST uses VQ autoencoders for content-adaptive training-free stylization (Gim, Yoo, and Uh 2024). Diffusion-prior stylization spans both trainable prompt-bank adaptation and training-free control: ArtBank learns an implicit style prompt bank on top of a pretrained diffusion model (Zhang et al. 2024), whereas StyleID and Z* manipulate diffusion attention at inference time without retraining (Chung, Hyun, and Heo 2024; Deng et al. 2024). SMS and StyleSSP further refine this large-prior stylization line through style-matching objectives and improved sampling initialization (Jiang et al. 2025; Xu et al. 2025). These methods can be visually strong, but they generally assume a style reference image and often inherit the inference cost or calibration issues of large generative priors. Our work instead targets compact retrainable latent integration with style-id conditioning rather than iterative diffusion sampling, prompt engineering, or per-style inference-time optimization.

Distribution matching and perceptual evaluation. Our objective is motivated by optimal transport and distribution matching. Sinkhorn distances and computational optimal transport provide efficient tools for approximate transport problems (Cuturi 2013; Peyré and Cuturi 2019), while the Schrodinger problem connects entropic transport to stochastic path measures (Léonard 2014). We use SWD as a lightweight terminal distribution objective and instantiate the reported Distinct5 headline rows as pairing-cache / terminal-SWD / kinetic OMF with adaptive projection routing through SA-SWD. Earlier exploratory variants in this line also studied OT/Sinkhorn endpoint assignment as a parent objective, and the formal analysis below refers to that machinery only where it is the actual object of study rather than the literal active Distinct5 training loop. Evaluation itself remains unsettled in arbitrary style transfer: prior studies report inconsistent protocols across papers and show that commonly used automatic metrics are style-dependent and require calibration to human preference (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our metric claim is therefore focused rather than universal. For separated art-to-art stress splits of this type, we make the unchanged-image prior explicit through an 'idt' reference and report Δ_{idt} so that target-style movement can be separated from preserved art-domain priors. In the Distinct5 results, the paper-facing ArtFID column is targetwise ArtFID; aggregate ArtFID is retained only as an auxiliary art-domain/identity diagnostic outside the main comparison table. We therefore combine perceptual and distributional measures including LPIPS (Zhang et al. 2018), FID (Heusel et al. 2017), KID (Bińkowski et al. 2018), DISTS (Ding et al. 2020), MUSIQ (Ke et al. 2021), MANIQA (Yang et al. 2022), and ArtFID (Wright and Ommer 2022). This broader metric set is important because raw CLIP-style and structural scores may fail to penalize grain-like artifacts or may partially reflect art-domain priors already present before stylization.

Method

Overview

LBM is a latent transport design in VAE space for domain-level artistic transfer. Let $z_0 \in \mathbb{R}^{4\times32\times32}$ denote the VAE latent of a content image and let s denote a target style domain identifier. It comprises four coupled components: (i) training-side endpoint pairing that selects candidate target latents without paired supervision; (ii) a time-conditioned transport field $v_\theta(z_t, t, s)$ that realizes the corresponding style-conditioned transport; (iii) kinetic regularization that discourages unnecessary latent displacement; and (iv) SA-SWD terminal matching at the integrated endpoint. At inference, the learned vector field is integrated with a fixed-step solver and decoded by the VAE decoder. In this decomposition, the tokenizer specifies the target region of style space, and the backbone is the executor that realizes the corresponding transport field. Earlier exploratory variants in the broader LBM line included OT-coupled parent objectives, but the reproduced Distinct5 headline rows use pairing-cache / terminal-SWD / kinetic OMF rather than active online Sinkhorn updates.

In the reproduced Distinct5 setting, the active headline rows use endpoint-mode OMF training rather than the historical random-time flow residual. Writing

$$\hat{v} = v_\theta(z_0, 1, s), \qquad \hat{z}_1 = z_0 + \hat{v}, \tag{1}$$

the active Distinct5 objective is

$$\mathcal{L}_{\mathrm{active}} = \lambda_{\mathrm{term}}\mathcal{L}_{\mathrm{term}}(\hat{z}_1) + \lambda_{\mathrm{kin}}\|\hat{v}\|_2^2, \tag{2}$$

where paired target latents for $\mathcal{L}_{\mathrm{term}}$ are selected by the pairing cache and the optional parent flow residual is disabled in the headline rows ($w_{\mathrm{flow}} = 0$). We therefore analyze LBM at the transport-design level and name objective-family differences only when they matter empirically.

Unpaired latent-domain sampling and endpoint pairing

We train directly on precomputed VAE latents without paired supervision. Each minibatch samples content latents

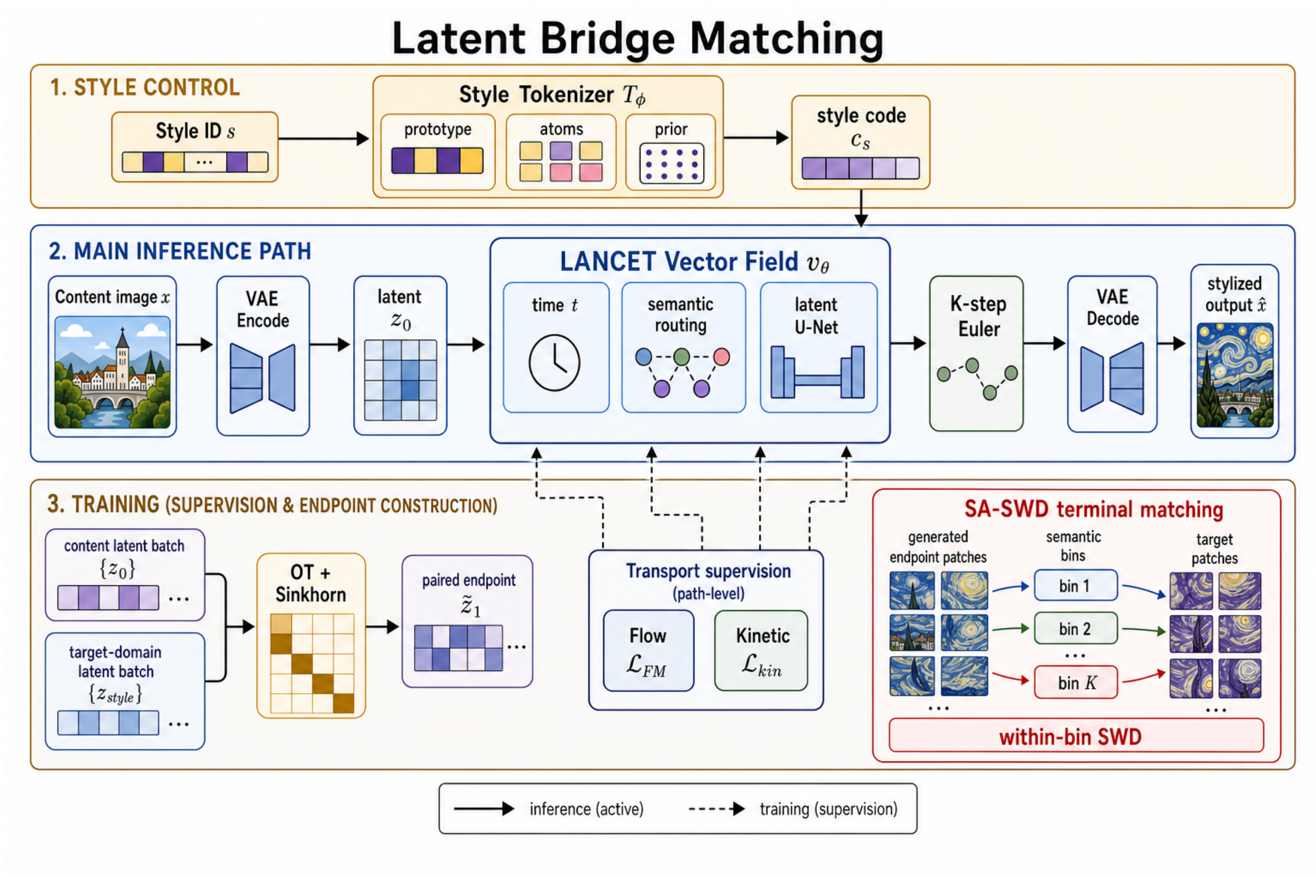


Figure 2: Overview of Latent Bridge Matching. The top band shows style-side control, where the style tokenizer maps the target style identifier into a compact style code. The middle band is the active inference path: content is encoded into a latent, passed through the LBM time-conditioned residual field with semantic routing, integrated by K-step Euler updates, and decoded into the stylized output. The bottom band summarizes training-only constraints: in the current Distinct5 mainline, target-domain latents enter through pairing-cache endpoint selection with endpoint-mode OMF training, while the broader bridge lineage also includes OT/Sinkhorn and flow-style parent variants. In the headline rows the parent flow residual is disabled ($w_{\text{flow}} = 0$); the active losses are terminal SA-SWD and endpoint kinetic regularization. The SA-SWD inset makes explicit how semantic bins define terminal within-bin SWD matching. Target-domain latents are therefore training-side supervision only, not inference-time conditioning inputs.

$\{z_c\}$ from all source domains and target-style latents $\{z_s\}$ from the requested target domain. Identity pairs are naturally included through uniform target-domain sampling, which stabilizes the learned transport around same-domain mappings.

In the reproduced Distinct5 mainline, candidate target endpoints are assigned by a prototype-aware pairing cache and selector schedule rather than by online Sinkhorn optimization at every training step. Concretely, for each content latent z_0 and target style s, the cache exposes a selector-restricted candidate set $\mathcal{C}(z_0, s) \subset \{z_s\}$, and the terminal target $\tilde{z}_1$ used by $\mathcal{L}_{\text{term}}$ is drawn from that scheduled set. For completeness, the broader bridge family also includes an OT-style mini-batch assignment diagnostic over sliced-Wasserstein costs. For a batch of content latents $\{z_c^{(i)}\}$ and target-style latents $\{z_s^{(j)}\}$, the SWD cost matrix is

$$C_{ij} = \mathrm{SWD}\big(z_c^{(i)}, z_s^{(j)}\big). \tag{3}$$

An entropic optimal-transport plan is then obtained via the Sinkhorn algorithm:

$$\pi^\star = \arg\min_{\pi} \sum_{ij} C_{ij}\pi_{ij} + \epsilon H(\pi), \tag{4}$$

where $H(\pi)$ is the entropy regularizer. Each content latent is matched to its coupled target $\tilde{z}_1 \sim \pi^\star(z_s|z_c)$. This historical parent objective motivates directional-coherent endpoint assignment without requiring spatially aligned targets, but it is not the literal active training loop for the Distinct5 F/H/K headline rows.

Time-conditioned latent transport supervision

The core of the method is a time-conditioned velocity field $v_\theta(z_t, t, s)$ that executes style-conditioned latent motion between selected training endpoints. The broader bridge lineage also studies a random-time flow-style parent objective, which we retain here only as family background because the active Distinct5 headline rows disable the corresponding flow residual. At a randomly sampled time $t \sim \mathcal{U}(0,1)$, the interpolated state is

$$z_t = (1-t)z_0 + t\tilde{z}_1 + \sigma\sqrt{t(1-t)}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{5}$$

and the target velocity is the derivative of the transport:

$$u_t = \tilde{z}_1 - z_0 + \sigma \frac{1-2t}{2\sqrt{t(1-t)}}\,\epsilon. \tag{6}$$

The parent transport objective supervises the network to predict this velocity:

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{z_0,\tilde{z}_1,t,\epsilon}\,\rho\big(v_\theta(z_t,t,s), u_t\big). \tag{7}$$

Here ρ is the pointwise velocity penalty. The implementation supports MSE, Huber, and L_1 variants in direct velocity-regression branches. The displayed flow objective above defines the historical parent transport objective; in the reproduced Distinct5 setting, training instead resolves to endpoint-mode OMF with terminal matching and kinetic regularization and keeps $w_{\mathrm{flow}} = 0$. In other words, the flow objective above is design-family background rather than the literal active headline loss for the Distinct5 rows.

Architecture. The time-conditioned backbone uses sinusoidal time embeddings added to the style code. The implementation is a U-Net-like latent trunk with four semantic cross-attention body blocks at 16×16 resolution. The target style is represented by a learnable spatial prior $P_s \in \mathbb{R}^{128\times16\times16}$. Semantic cross-attention injects the style prior into content features, while skip connections preserve spatial structure.

Style tokenizer as executable control

The style-side interface is deliberately separated from the latent renderer. In the main domain-level protocol, the tokenizer receives only a target style identifier and learned style parameters available at inference:

$$c_s = T_\phi(s), \qquad v_\theta = F_\theta(z_t, t, c_s). \tag{8}$$

It does not read a target image or target latent from the current evaluation batch. Target-domain latents remain on the loss side of the graph through training-side endpoint pairing and SA-SWD, but they are not conditioning evidence for T_ϕ. This boundary is important: otherwise training would solve a reference-guided problem while evaluation reports a style-id problem. In this sense, T_ϕ names where transport should go, while F_θ decides how much latent motion is actually needed to get there. Throughout this paper, "style tokenizer" means a style-id-conditioned control map, not an image-token or diffusion tokenizer and not a target-image encoder.

The simplest tokenizer is a per-style code table. Our current implementation generalizes this interface without changing the renderer input contract:

$$c_s = p_s + \gamma \sum_{k=1}^{K} \alpha_{s,k} a_k, \qquad \alpha_s = \mathrm{softmax}(\ell_s/\tau), \tag{9}$$

where p_s is an optional direct prototype, $\{a_k\}$ are shared style atoms, and γ controls the residual atom contribution. In the current mainline, this interface is paired with a prototype-aware latent target queue so that the renderer receives a stable style carrier without changing its single-code input contract. The tokenizer therefore exposes an executable style vocabulary while preserving the compact interface required by the existing renderer.

For execution, the backbone may also use style spatial priors and content-conditioned routing. These modules modulate where a target-style control is applied, not what target style is requested. This distinction matches the empirical diagnosis: the key question is not raw tokenizer capacity in isolation, but whether the style carrier remains executable after it passes through a content-conditioned renderer. Prototype-aware target queues and content-guided routing improve that executed style-content frontier, which is why we phrase tokenizer evidence at the level of *executable representation*: code geometry matters only insofar as generated latent residuals remain distinguishable after the renderer consumes the code.

Complementary roles. The components address distinct subproblems. Training-side endpoint pairing provides candidate target endpoints for unpaired supervision, but not transport dynamics; the broader OT diagnostic explains one historical route to such assignments. Transport supervision learns how to move toward those endpoints, but by itself does not guarantee that the integrated endpoint matches target-domain texture statistics. SA-SWD regularizes this endpoint mismatch, while semantic cross-attention provides local routing for where style is written and supplies a routing-derived heuristic for projection selection. The destructive ablations directly verify the roles of terminal distribution matching and kinetic regularization. On the current Distinct5 packet, the dominant gain comes from distributional terminal matching itself, which avoids pointwise latent collapse, while semantic routing remains a tested axis-selection heuristic rather than a separately proven primary driver.

Adaptive-projection terminal distribution matching

In addition to transport supervision, a terminal loss regularizes the Euler-integrated endpoint toward the target style distribution. Let $\Phi_\theta(z_0, s)$ denote the endpoint obtained by integrating v_θ from z_0 with style s. The terminal loss is

$$\mathcal{L}_{\mathrm{term}} = \mathrm{SWD}_1\big(\Phi_\theta(z_0,s),\ \tilde{z}_1\big), \tag{10}$$

where the SWD compares generated endpoint patches with coupled target patches. We use semantic cues to choose SA-SWD projection directions. Let K_s denote the keys produced by semantic cross-attention for target style s. We rank spatial target positions by the average magnitude of K_s and use the corresponding target latent patch vectors as normalized projection directions. This defines the SA-SWD instantiation used in the current mainline: terminal distribution matching is still computed by sorted one-dimensional projections, but the projection axes are adaptively chosen from the same routing signal that injects style into the backbone.

Across the broader LBM family, the training objective can be written as

$$\mathcal{L} = \mathcal{L}_{\text{transport}} + \lambda_{\text{term}}\mathcal{L}_{\text{term}} + \lambda_{\text{kin}}\mathcal{L}_{\text{kin}}, \tag{11}$$

where $\mathcal{L}_{\text{transport}}$ denotes the optional parent transport residual retained only for family-level completeness. In the current Distinct5 headline rows this term is disabled ($\mathcal{L}_{\text{transport}} = 0$, $w_{\text{flow}} = 0$), so the active loss reduces to Eq. (2). The kinetic loss $\mathcal{L}_{\text{kin}} = \mathbb{E}\|v_\theta\|_2^2$ penalizes excessive latent displacement. The primary configuration uses $\lambda_{\text{term}} = 20.0$ and $\lambda_{\text{kin}} = 1.0$. Color, cycle, NCE, and repulsive losses are disabled in the main setting. This decomposition separates family-level template terms from the losses that are actually active in the Distinct5 headline regime.

Metric choice in latent space. The training objective does not treat all latent discrepancies as Euclidean reconstruction errors. The kinetic term uses an L_2 penalty on velocity because it represents path energy, while terminal endpoint matching uses Wasserstein-1 style discrepancies over sorted projections. This separation is deliberate in compressed VAE latents: pointwise endpoint MSE would reward mean solutions and over-penalize semantically valid local rearrangements, whereas SA-SWD preserves distributional supervision without requiring paired spatial correspondence. The current reproduced Distinct5 checkpoints therefore place style pressure on training-side endpoint pairing plus W_1-style terminal matching rather than on latent endpoint MSE. In the measured Distinct5 regime, this design is consistent with repaired endpoint-only controls underperforming the current mainline, without claiming a universal theorem against pointwise endpoint losses.

Theoretical grounding of latent transport. Section provides bounded design-grounding results. Theorem 1 connects the training kinetic loss to the inference-time path action, and Theorem 2 bounds Euler discretization error at $O(1/K)$ for the active few-step field/solver regime used in the current Distinct5 mainline. Theorem 3 concerns a historical OT-style endpoint-assignment variant from the broader LBM family and is included only as family-level background.

More concretely, let $\mathcal{P}_r(\hat{z}_1)$ be the set of latent patches of size r extracted from the generated endpoint, and let $\mathcal{P}_r(z_s)$ be the corresponding patch set from target-style samples. For projection directions $\{\theta_m\}_{m=1}^M$, the empirical terminal objective is

$$\mathcal{L}_{\text{swd}} = \sum_{r\in\mathcal{R}} \frac{1}{M} \sum_{m=1}^{M} \Big\| \text{sort}\big(\mathcal{P}_r(\hat{z}_1)\theta_m\big) - \text{sort}\big(\mathcal{P}_r(z_s)\theta_m\big) \Big\|_1. \tag{12}$$

Here $\mathcal{R} = \{3, 5, 7, 15\}$ in the primary configuration. The loss therefore asks whether the generated endpoint has target-style patch statistics along semantic routing directions, not whether it matches any particular artwork spatially.

Formal analysis

We provide bounded design-grounding results rather than a single unified theorem for every training variant. Theorem 1 and Theorem 2 analyze the active few-step field/kinetic regime used in the current Distinct5 mainline, while Theorem 3 is retained only as historical parent-background completeness for an OT-style endpoint-assignment variant from the broader LBM family. Full proofs appear in the supplement. For brevity we state the results and their empirical support.

Notation. Recall the OMF training regime: $\hat{v} = v_\theta(z_0, 1, s)$, $\hat{z}_1 = z_0 + \hat{v}$. At inference, integration follows $dz/dt = v_\theta(z, t, s)$ with K-step Euler: $z_{k+1} = z_k + \Delta t\, v_\theta(z_k, t_k, s)$ where $\Delta t = 1/K$, $t_k = (k + 0.5)\Delta t$. Define the path action $\mathcal{A}(v) = \int_0^1 \mathbb{E}\|v_\theta(z(t), t, s)\|^2\, dt$ and the training kinetic loss $\mathcal{L}_{\text{kin}} = \mathbb{E}\|v_\theta(z_0, 1, s)\|^2$.

Design interpretation. Under bounded temporal drift and Lipschitz continuity, endpoint kinetic regularization acts as a tractable proxy for inference-time path action. Theorem 1 formalizes why a lightweight endpoint penalty can act as a practical proxy for trajectory straightness in the operating regime studied here, without requiring expensive multi-step supervision.

Theorem 1 (Kinetic regularization as a path-energy proxy). Assume that for a fixed content z_0 the velocity norm varies at most δ with t, i.e., $\sup_t \big| \|v_\theta(z_0, t, s)\| - \|v_\theta(z_0, 1, s)\| \big| \le \delta$, and that v_θ is L-Lipschitz in z. Let M bound $\|v_\theta\|$ along the trajectory. Then the path action decomposes as

$$\mathcal{A}(v) = \mathcal{L}_{\text{kin}} + \varepsilon_t + \varepsilon_z, \tag{13}$$

where $|\varepsilon_t| \le 2\delta\sqrt{\mathcal{L}_{\text{kin}}} + \delta^2$ and $|\varepsilon_z| \le LM$. If δ and LM are small, the training kinetic loss provides a tight proxy for the inference-time path energy.

Empirical support. On a historical D0 probe consistent with the current few-step regime (128-step integration): $\mathcal{L}_{\text{kin}} \approx 1017$, $\mathcal{A}(v) \approx 698$, yielding $|\varepsilon_t + \varepsilon_z| \approx 319$. Removing $\mathcal{L}_{\text{kin}}$ (D2) reduces the path length ratio from 0.98 to 0.94, confirming the kinetic term controls path regularity.

Theorem 2 (Euler discretization bound). Under the L-Lipschitz condition on v_θ, the K-step Euler integration error satisfies

$$\|z(1) - z^{(K)}\| \le \frac{C}{K}, \qquad C = \frac{1}{2}(ML_t + M^2 L)\,\frac{e^L - 1}{L}, \tag{14}$$

where L_t bounds the t-derivative of v_θ.

Empirical support. Measured over 160 content latents (D0 model, 256-step reference): $\|z^{(K)} - z^{(256)}\| = \{10.7, 3.80, 1.74, 0.84, 0.40, 0.19, 0.084, 0.027\}$ for $K = 1, 2, 4, 8, 16, 32, 64, 128$; the error halves when K doubles, confirming the $O(1/K)$ scaling. Because VAE latents exhibit non-trivial local curvature, few-step solvers can otherwise accumulate visible drift. Here the measured $O(1/K)$ decay is consistent with the expected solver trend on this historical D0 probe. At $K = 12$ in that probe, the error is < 0.6 relative to the 256-step reference.

Historical parent theorem (Theorem 3). This result concerns the historical OT-style parent variant rather than the literal active Distinct5 headline loop. Let the SWD cost $c(\cdot,\cdot)$ be L_c-Lipschitz and let $\pi^\star$ be the ε-Sinkhorn plan on a batch of size B. Let D denote the diameter of the target latent set. Then, for any two content latents z_0, z_0', the expected matched targets satisfy

$$\left\|\mathbb{E}_{\pi^\star}[\,\tilde{z}_1 \mid z_0\,] - \mathbb{E}_{\pi^\star}[\,\tilde{z}_1 \mid z_0'\,]\right\| \leq \frac{D \cdot L_c}{\varepsilon}\,\|z_0 - z_0'\| + O(\varepsilon). \tag{15}$$

Thus the OT assignment map is (DL_c/ε)-Lipschitz: nearby content latents receive nearby matched targets. Random uniform coupling provides no such guarantee. The Lipschitz property implies smoother supervision gradients and more directionally coherent displacement vectors across the batch.

Empirical support. Compared over 640 content latents: OT coupling yields mean cosine similarity to the batch-mean direction of 0.150, vs. 0.124 for random coupling, a 21% improvement. The transport cost is also reduced by 4.2% (0.119 vs. 0.124).

Experiments

Protocol hierarchy and metrics

We report two non-interchangeable result families: Distinct5-512 as the main stress benchmark, and the historical strict-750 packet as contextual support. They are separated because dataset, resolution, and protocol changes can alter the meaning of the CLIP-style / LPIPS trade-off.

Historical strict-750. The legacy packet uses photo, Hayao, Monet, Van Gogh, and Cezanne, with all 5×5 source-target directions and 30 source images per source domain, for 750 outputs. It is used for comparisons to SaMST, S2WAT, StyleID, AdaIN, StyTr2, CAST, AesFA, and AesPA-Net.

Distinct5-512 stress split. The stress split uses Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 1000 training images and 30 held-out test images per class. These classes were selected by CLIP-style separation over WikiArt categories rather than hand-picked for LBM. Evaluation again uses all 5×5 directions over 750 images.

We report CLIP-style (CLIP-S), CLIP-content (CLIP-C), LPIPS, and

$$\mathrm{EC} = \text{CLIP-style} \cdot (1 - \mathrm{LPIPS}). \tag{16}$$

For Distinct5-512 we additionally report style gain over the unchanged-image reference,

$$\Delta_{\mathrm{idt}} = \text{CLIP-S(method)} - \text{CLIP-S(idt)}, \tag{17}$$

under both the full 5×5 protocol and the transfer-only off-diagonal subset. On this split, a method is only considered successful if it exceeds the unchanged-image reference without moving into a clearly damaged LPIPS / artifact regime. We treat CLIP-S as target-style affinity, LPIPS as content displacement, targetwise ArtFID as a plausibility / artifact diagnostic, and EC only as a compact summary. For artifact-sensitive analysis we also report MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, shallow Gram micro statistics, and KID where available. All baselines use the same source images, target domains, preprocessing, and identity directions. Arbitrary-style baselines receive one reference image per target domain, whereas LBM receives only the target domain identifier.

Table 1: Distinct5-512 all-pairs results on the shared 5×5 / 750-output protocol. 'idt' is the unchanged-image control, and 'tw-ArtFID' denotes targetwise ArtFID. SaMAM is reported only at the trusted 2250-step checkpoint. Timing is summarized separately in Table 3.

Method	CLIP-S↑	LPIPS↓	EC↑	tw-ArtFID↓	Δ_{idt} ↑	$\Delta_{\mathrm{idt,tr}}$ ↑
idt (no-op)	0.6801	0.0000	0.6801	216.5	0.0000	0.0000
SaMAM 2250	0.5811	0.3538	0.3755	146.1	-0.0990	-0.0877
SaMST-512 e15	**0.7247**	0.6255	0.2714	395.7	**+0.0446**	**+0.0558**
LBM-F e1	0.6969	**0.3186**	**0.4748**	**122.6**	+0.0168	+0.0244
LBM-K e1	0.7010	0.3623	0.4470	157.2	+0.0209	+0.0312

Table 2: Representative Distinct5 transfer-only pairs. Cells report CLIP-S / LPIPS. ER = Early Renaissance, Imp = Impressionism, Min = Minimalism, Roc = Rococo, Ukiyo = Ukiyo-e.

Pair	No-op	SaMAM-2250	SaMST e15	LBM-F e1
ER→Min	0.607	0.566 / 0.387	0.863 / 0.644	0.710 / 0.498
ER→Ukiyo	0.663	0.571 / 0.321	0.748 / 0.195	0.724 / 0.396
Imp→Min	0.602	0.623 / 0.585	0.892 / 0.731	0.823 / 0.483
Roc→Ukiyo	0.515	0.539 / 0.542	0.582 / 0.604	0.599 / 0.295

Distinct5-512 stress benchmark

Distinct5-512 is designed so that unchanged output is not a plausible success proxy: the *idt* baseline already reaches CLIP-S 0.6801. Signed gain over idt is therefore the first sanity check. LBM-F reaches 0.6969 / 0.3186 after 1.2 minutes of training, and LBM-K reaches 0.7010 / 0.3623 while staying in the same low-damage region. SaMST-512 pushes raw style to 0.7247, but only at LPIPS 0.6255 and targetwise ArtFID 395.7. SaMAM-2250 stays below the idt floor despite lower ArtFID. Table 2 shows the same pattern on representative directions: useful transfer must exceed idt, not merely lower ArtFID or raise absolute CLIP-S.

Table 3 reports the closed same-cost rows, and Table 2 shows representative transfer directions.

A paired bootstrap over the 600 transfer rows gives 95% intervals of $[0.0210, 0.0280]$ for LBM-F and $[0.0273, 0.0352]$ for LBM-K, confirming stable positive transfer-only Δ_{idt} on this split. SaMST also stays above IDT in the bootstrap, but only in the same high-LPIPS regime.

Table 3: Distinct5-512 matched same-cost records under the reviewed full 5×5 / 750-output protocol. Train time excludes evaluation; inference is pure 750-image generation wall time.

Method	Recorded point	Train to point	Inference (750 imgs)	ms/img
LBM	step 350	**1.9 min**	**84.4 s**	**113**
SaMAM	step 16	2.2 min	109.2 s	146
SaMST	40-step/style	2.0 min	345.7 s	461

Note. All three methods were retrained to roughly the same two-minute budget before evaluation. This table reports the matched-budget row, not the strongest retained LBM frontier point. The stronger LBM frontier rows still appear earlier at about 1.2 minutes in the page-1 teaser. Peak local GPU memory is about 5.0 GB for LBM, 6.1 GB for SaMST, and 7.5 GB for SaMAM.

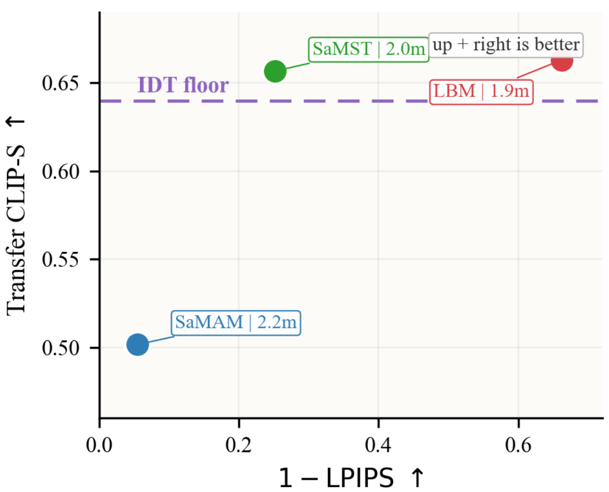


Figure 3: Matched same-cost view on Distinct5-512. Transfer CLIP-S is plotted against $1 - \text{LPIPS}$; the dashed line marks the transfer-only IDT floor.

Adaptation cost and accessibility. Table 3 and Figure 3 summarize the matched same-cost rerun. After about two minutes of training, the LBM step-350 row reaches CLIP-S 0.6629 at LPIPS 0.3377 and stays +0.0230 above the Distinct5 IDT floor. Same-cost SaMST is only +0.0166 and far more destructive (LPIPS 0.7487, 461 ms/image), while same-cost SaMAM remains below IDT at -0.1383 after 2.2 minutes. These are matched-budget checkpoints, not the best retained frontier points: the stronger LBM-F/K e1 rows still appear on page 1. All three runs stay within the same 8 GB RTX 4070 laptop envelope, peaking at about 5.0/6.1/7.5 GB for LBM/SaMST/SaMAM. The practical claim is therefore narrow but clean: within the same minute-scale budget, compact latent transport reaches useful positive transfer without large-prior adaptation.

Historical strict-750 comparison

We use the historical strict-750 packet as contextual support. Table 4 places LBM in a favorable low-damage region: StyleID has the highest raw CLIP-S but collapses content, and SaMST is slightly higher on CLIP-S / CLIP-C while LBM is better on LPIPS, EC, and the artifact metrics below. We treat this as bounded historical support rather than a universal same-interface ranking.

Historical cost evidence is intentionally conservative. The only closed row we retain is LBM: 5.2 minutes to train and 114 ms/image on strict-750. Other baselines are omitted because their train/infer logs are mixed-protocol or incomplete.

Artifact-sensitive diagnosis

The artifact difference is clearest on the historical 256-resolution packet, where SaMST is the closest recovered efficient baseline. Figure 4 shows the whole-image contrast; Table 6 then pairs matched crops with artifact-sensitive metrics on the same packet.

Table 6 shows that LBM is not winning by doing less. Relative to SaMST, it improves MUSIQ (36.10 $\rightarrow$ 49.21), MANIQA (0.314 $\rightarrow$ 0.406), DISTS-content (0.294 $\rightarrow$ 0.248), HF-Patch-KID (6.76 $\rightarrow$ 4.17), and FFT slope error (1.054 $\rightarrow$ 0.547). A paired bootstrap over the 750 per-image scores gives 95% intervals of $[0.4467, 0.4563]$ for LPIPS and $[0.7091, 0.7234]$ for CLIP-S.

Tokenizer and representation ablations

These ablations point to control, not capacity. More codebook capacity alone does little, content-guided routing helps, the prototype-aware queue gives the largest gain, and content-adaptive routing buys more style at the cost of larger motion. The next interface should therefore separate target-style carrying from content-risk control.

Non-training probes support the same boundary. Tokenizer codes remain nearly full-rank, but executed residuals are much more aligned and strongly source dominated. On matched Distinct5 probes, executor-side refresh helps more than style-side refresh alone. The bottleneck is therefore execution, not simple code separability.

Mechanism ablations

Destructive ablations show the intended trade-off. Removing SA-SWD preserves content but weakens style; removing kinetic regularization preserves style but collapses content; strong color loss hurts the balance. Path metrics give the same reading: the full model has PLR 0.98, w/o SA-SWD rises to 0.9995, and w/o kinetic drops to 0.94 with content collapse. A matched same-family Distinct5 H packet supports the same bounded claim: weaker kinetic regularization increases endpoint/path/peak motion and worsens CLIP-S / LPIPS.

Sensitivity and low-cost adaptation

The category-weight sweep did not beat balanced sampling consistently. The best EC point is K2 balanced default at epoch 3; the best raw-style point is K1 balanced default at epoch 8. We therefore keep this sweep as supporting sensitivity rather than main evidence.

Step-count sensitivity is also stable. Moving from 1 to 16 integration steps changes all-pairs CLIP-S only from 0.7159

Table 4: Historical strict-750 comparison in grouped family format. Higher is better for CLIP-S, CLIP-C, and EC; lower is better for LPIPS. Artifact-sensitive metrics are reported separately in Table 6.

Metric	CNN-based			Transformer-based			Training-free		Latent transport
	AesPA	AesFA	AdaIN	StyTr2	S2WAT	SaMST	CAST	StyleID	LBM
CLIP-S ↑	0.5976	0.6496	0.7130	0.6374	0.7139	0.7194	0.6652	**0.7597**	0.7161
CLIP-C ↑	0.5009	0.5497	0.6990	0.6001	0.7465	**0.8193**	0.5562	0.5519	0.8086
LPIPS ↓	0.8215	0.7392	0.6298	0.6345	0.5263	0.4664	0.7263	0.7497	**0.4514**
EC ↑	0.1067	0.1694	0.2639	0.2330	0.3382	0.3839	0.1821	0.1902	**0.3928**

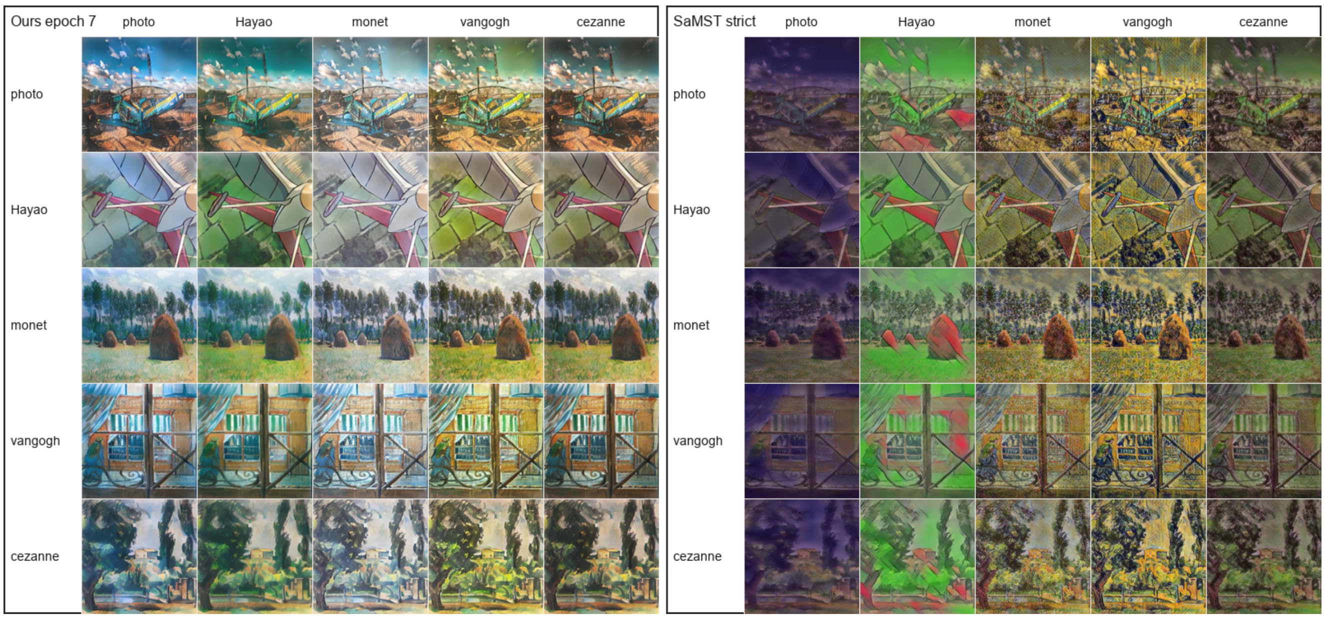


Figure 4: Historical 256-resolution LBM vs. SaMST grids. SaMST often pushes more texture but adds muddy grain, color spill, and broken edges; LBM stays cleaner.

Table 5: Historical 256×256 standard-benchmark operating-point cost. Training time is the wall-clock cost to the manuscript operating point; inference is the end-to-end runtime for the 750-image protocol. This is recorded operating-point context, not a normalized time-to-quality comparison.

Method	Train to point	Inference (750 imgs)	ms/img
LBM	**5.2 min**	85.4 s	114

Note. The LBM row is a preserved timing record. Other historical rows are intentionally omitted here because the available train/infer evidence is either probe-estimated, mixed-protocol, or incomplete rather than a closed operating-point record.

to 0.7162, while LPIPS stays near 0.4514. Four steps already recover near-identical quality, with strict-750 throughput rising from 8.43 to 9.54 images/s. A 3×3 sweep over λ_{kin} and λ_{swd} shows the expected smooth trade-off: stronger kinetic weight improves content, while stronger terminal pressure increases style at the cost of LPIPS.

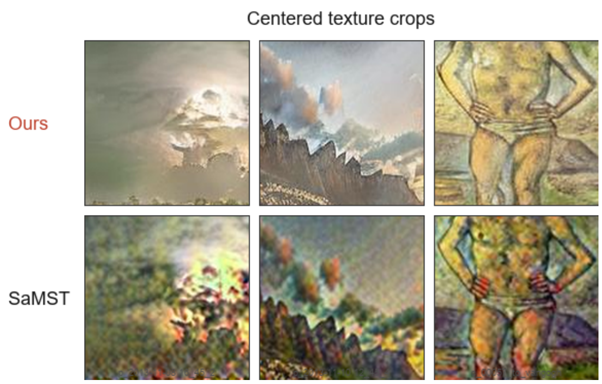
All reported evaluations use fixed seeds, stored resolved configurations, source snapshots, checkpoints, and per-epoch logs. Historical timings were measured on an RTX 4070 Laptop GPU unless otherwise noted. Distinct5 same-cost evidence is summarized by Table 3 and Figure 3; historical cost is reported in Table 5. Across both packets, the takeaway is practical: meaningful style-id customization can remain in a minute-to-hour consumer-GPU regime without adapting a large pretrained generator.

Discussion and Limitations

The main lesson is that style transfer quality cannot be inferred from raw CLIP-style alone. On the historical strict-750 protocol, StyleID has the highest raw style but collapses content, SaMST is slightly stronger than LBM on CLIP-S and CLIP-C but shows grain-like local artifacts, and LBM occupies a cleaner LPIPS/artifact-sensitive operating point. Distinct5-512 sharpens the same lesson further because its no-op baseline is measured explicitly: higher raw style is achievable, but only by moving into a severe LPIPS and targetwise ArtFID failure region, and some baselines fail even to exceed the identical-image prior. This exposes a community-relevant blind spot in art-to-art evaluation: preserved art-domain priors can masquerade as target-style suc-

Table 6: Artifact-sensitive diagnosis on the historical strict-750 packet. Top: metrics. Bottom: matched texture crops from the same rows as Figure 4.

Metric	Ours e7	SaMST	S2WAT
MUSIQ ↑	49.2059	36.0950	36.5256
MANIQA ↑	0.4057	0.3139	0.1754
DISTS-content ↓	0.2477	0.2943	0.2942
HF-Patch-KID ↓	4.1694	6.7598	12.6623
FFT slope error ↓	0.5473	1.0536	0.7017
Gram micro ↓	0.0798	0.0947	0.1085



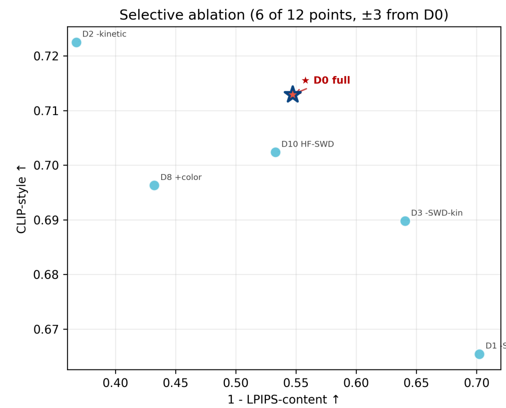


Figure 5: Destructive ablations: w/o SA-SWD weakens style; w/o kinetic collapses content.

cess unless the unchanged-image floor is reported. Earlier AST evaluation work already notes that protocols are inconsistent across papers and that automatic scores can be style-dependent or misaligned with human judgment (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our claim is therefore concrete but local: on Distinct5-512, the unchanged-image reference and Δ_{idt} act as useful diagnostics because they agree with the representative-direction sanity check in Table 2 and with the observed failure of sub-IDT baselines to move consistently toward the requested target style. The practical implication is separate from that metric point: the same experiments show that meaningful style-id customization can be trained and deployed in a lightweight small-model regime, lowering the barrier for researchers and creators who want customized stylization without maintaining a large generative backbone. We therefore treat adaptation cost as community-relevant evidence rather than an appendix-side detail and encourage future domain-style papers to report it alongside quality metrics. Across both protocols, the conclusion is consistent: on Distinct5-512 and within the currently reproduced positive-Δ_{idt} operating points, LBM occupies a favorable measured CLIP-S / LPIPS region, while the remaining open mechanism question is how faithfully style-side distinctions survive execution rather than whether code-space changes can be induced at all.

The tokenizer experiments narrow the design question. Within the tested Distinct5 design space, raw code capacity alone is not the bottleneck. The sharper issue is whether target-style information survives execution through a content-conditioned vector field strongly enough to yield real style gain beyond the unchanged-image reference. The current evidence therefore favors target-style carriers, prototype-aware target queues, and content-aware routing over undifferentiated token expansion. Matched successor probes reinforce the same point: tokenizer code separation maps only moderately into executed separation, while style gain over the unchanged-image reference tracks executed movement more closely than raw code geometry. This keeps executed style survival as the sharper mechanism question in the current successor family, and it suggests that the next representation stage should factor target representation from execution budget rather than simply adding parameters.

The study has several limitations. First, the main protocol is domain-level rather than reference-image-specific, so it does not attempt to reproduce a unique style image. Second, Euler integration is deliberately simple; although the step sweep is stable in the reported regime, more aggressive paths or higher resolutions may need better solvers or stronger constraints. Third, SA-SWD is an adaptive-projection distribution loss; current controls show that most of its benefit comes from the distributional term rather than from the specific semantic projection rule. Fourth, artifact-sensitive metrics are still proxies for visual preference; human studies or rubric-based VLM assessment would strengthen the claim further. Finally, Distinct5-512 is a five-domain stress split rather than a universal art benchmark, so broader domain coverage remains future work even though the current cross-method comparison is already informative on this split; we therefore do not claim universal adoption of IDT-style diagnostics without broader validation.

Conclusion

We presented Latent Bridge Matching, a compact latent transport design for domain-level artistic transfer. The method separates training-side endpoint selection, path-wise field learning, adaptive-projection terminal distribution matching, and style-tokenized execution. Under the reproduced historical strict-750 protocol, LBM delivers a favorable reproduced balance of style strength, content fidelity, artifact control, and practical footprint within the mixed-family comparison packet considered here. The Distinct5-

Table 7: Distinct5-512 tokenizer ablations. Control quality depends more on target selection and routing than on raw code capacity.

Variant	Best epoch	CLIP-S	LPIPS	Interpretation
Direct atom residual baseline	8/1	0.6876	0.4468	weak baseline; simple residual atoms do not solve representation
Class-local prototypes	8/1	0.6849	0.4464	more per-class capacity, no better style geometry
Global VQ atoms	8	0.6873	0.4446	small LPIPS gain, no style breakthrough
Content-guided spatial routing	2	0.6907	0.4226	first simultaneous style and LPIPS gain
VQ + content-guided routing	1	0.6898	0.4156	stronger content preservation, style still limited
Prototype-aware latent target queue	1/3	0.6973	0.3331	major jump from better internal target distribution
Annealed prototype queue (F)	1	0.6969	**0.3186**	current LPIPS reference point
Hard-explore prototype queue (H)	2/1	0.6994	0.3213	current balanced point
Content-adaptive VQ atom routing (K)	1	**0.7010**	0.3623	style boost, but larger endpoint movement

Table 8: Destructive ablations under the historical strict-750 protocol.

Variant	CLIP-S	CLIP-C	LPIPS	EC
D0 full	0.7014	0.8022	0.4593	0.3791
D1 w/o SA-SWD	0.6708	0.8989	0.3490	0.4368
D2 w/o kinetic	0.7159	0.6624	0.6375	0.2596
D8 strong color loss	0.6923	0.6629	0.5675	0.2994

512 stress benchmark sharpens the paper's main evaluation point: once the identical-image prior is made explicit, LBM stays above the no-op floor and occupies a favorable positive-Δ_{idt} CLIP-S / LPIPS region among the reproduced compact operating points, while lower ArtFID alone does not guarantee useful target-style movement. On this split, reporting the IDT floor together with Δ_{idt} is a useful diagnostic rather than an optional extra statistic. The matched same-cost rerun makes the practical point explicit: under roughly two-minute training budgets, LBM reaches CLIP-S 0.6629 at LPIPS 0.3377 and remains +0.0230 above IDT, whereas same-cost SaMST reaches only +0.0166 at LPIPS 0.7487 and same-cost SaMAM remains below IDT. Just as importantly, the reproduced minute-scale RTX 3060 adaptation path and the 3.91M / 114 ms historical operating point show that customized stylization can remain a small-model learning problem rather than a large-prior adaptation problem, lowering the barrier for researchers and creators who want low-cost style-specific models on consumer GPUs. The tokenizer study then isolates the remaining mechanism question. Within the tested Distinct5 tokenizer variants, matched localization controls show that executor-side refresh recovers more than style-side refresh alone on the current local surface, while the same-family H path-stability packet supports the bounded claim that kinetic regularization acts as a practical stabilizer in the current OMF / field regime. The next representation target is therefore to preserve target-style separation through execution itself, via explicit target carriers and tighter content-risk control.

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Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: **Yes**.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: **Yes**.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: **Yes**.

Does this paper make theoretical contributions? Yes (design-grounding). The paper provides bounded formal analysis with proofs and empirical checks: Theorem 1 and Theorem 2 analyze the active field/solver regime through a path-action decomposition for kinetic regularization and an $O(1/K)$ Euler discretization bound, while Theorem 3 analyzes a historical OT-style assignment variant as family-level background. Full proofs are in the supplement. These results justify the three-part design without claiming equivalence to a full Schrödinger Bridge solver.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: **Yes**.
- All novel datasets introduced in this paper are included in a data appendix: **NA**.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **NA**.
- All datasets drawn from the existing literature are accompanied by appropriate citations: **Yes**.
- All datasets drawn from the existing literature are publicly available: **Yes**.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing: **NA**.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: **Partial**.
- Any code required for pre-processing data is included in the appendix: **No**.
- All source code required for conducting and analyzing the experiments is included in a code appendix: **No**.
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **Partial**.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: **Partial**.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: **Partial**.
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: **Partial**.
- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: **Yes**.
- This paper states the number of algorithm runs used to compute each reported result: **Yes**.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: **Yes**.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: **Partial** (selected confidence intervals and bootstrap summaries are reported, but not every manuscript comparison is presented as a closed significance claim).
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: **Partial**.